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**NAAC
GRADE A+**
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Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 2

Title: Creating Power BI Visualizations to Analyze Sales Performance Across Products, Categories, and Regions

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Creating Power BI Visualizations to Analyze Sales Performance Across Products, Categories, and Regions

GitHub Repository Link: https://github.com/nidhidev2122/Business_analytics

Problem Statement

Management wants to quickly understand sales performance across products, categories, and regions. Create appropriate Power BI visualizations to identify key business patterns.

Dataset used: Sample Superstore

Source: <https://www.kaggle.com/datasets/naveenkumar20bps1137/sample-superstore>

Methodology / Steps Followed

Step 1: Data Collection

The Sample Superstore dataset was used for this experiment. The dataset was sourced from Kaggle and downloaded as a single CSV file

Total Records: 9,994 rows

Total Columns: 18 (after removing Row ID)

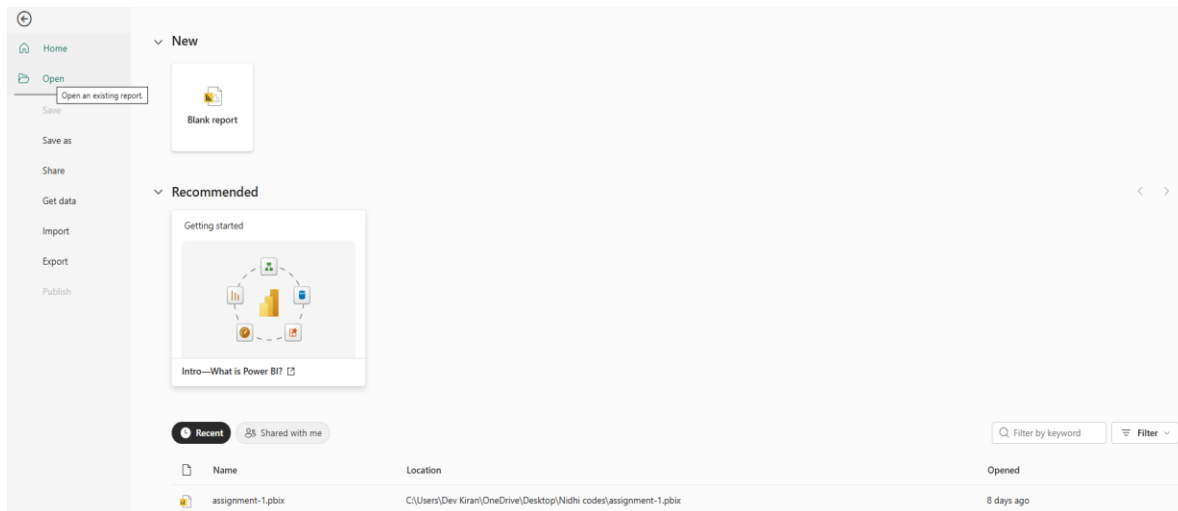


Figure 1: Power BI Desktop Home Screen

Step 2: Data Import and Transformation in Power BI

Power BI Desktop was opened and the dataset was imported using the Get Data option by selecting Text/CSV connector. The file Sample_Superstore.csv was loaded directly as a flat file.

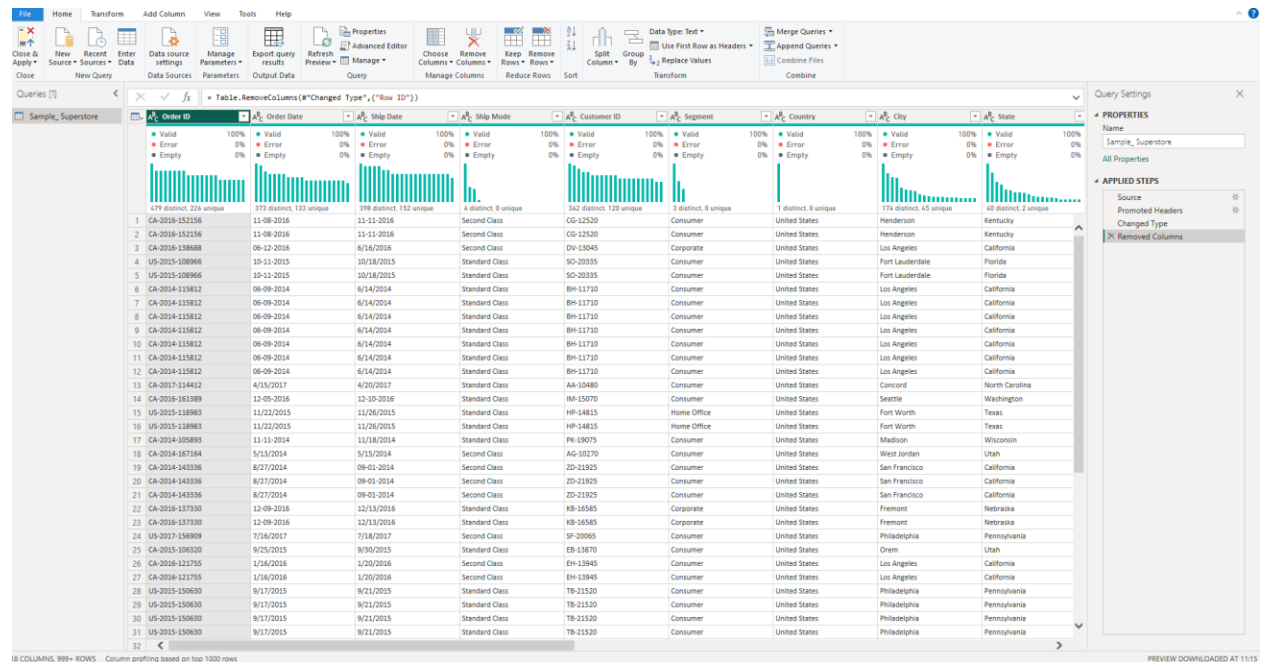


Figure 2: Power Query Editor

Step 3: Dashboard Development

An interactive dashboard titled "Superstore Sales Performance Dashboard" was developed in the Report View.

Output

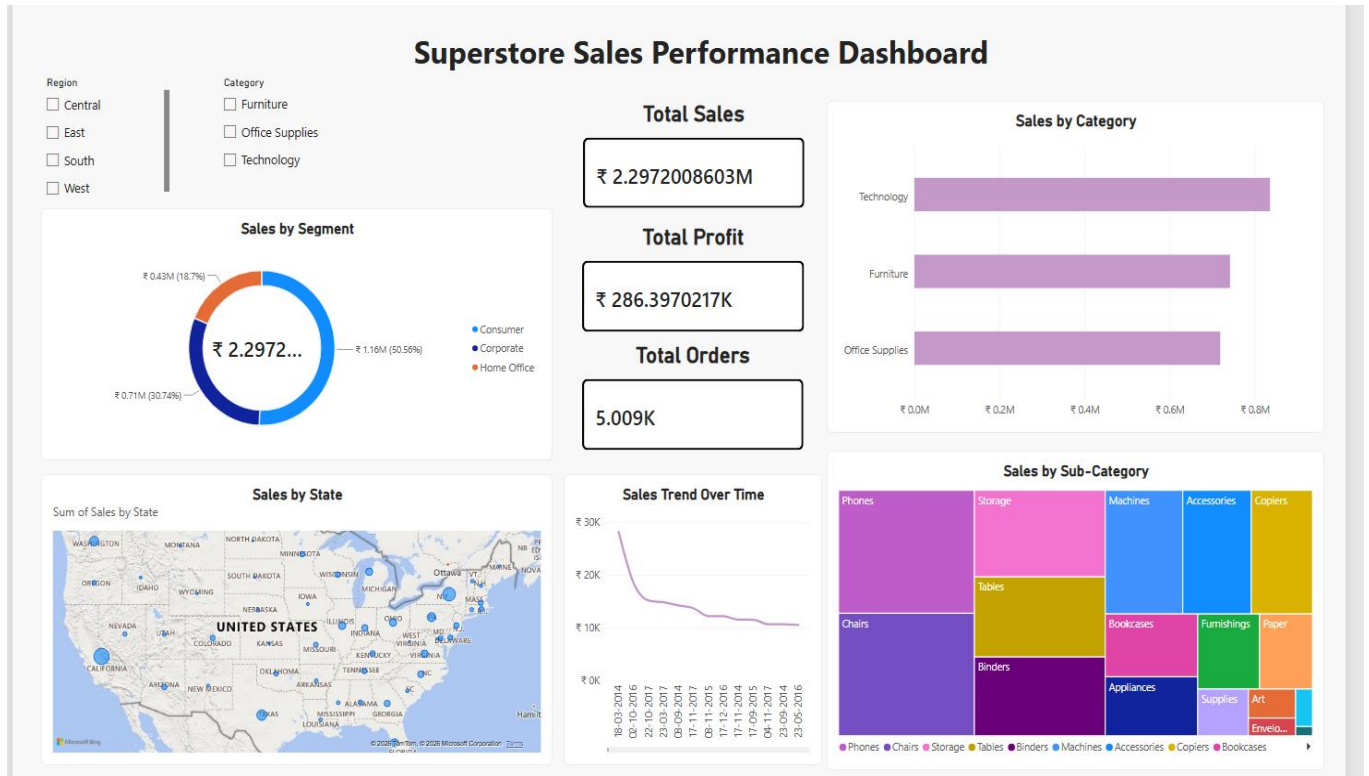


Figure 3: Superstore Sales Performance Dashboard — Report View

The final Power BI dashboard titled "Superstore Sales Performance Dashboard" successfully visualizes the retail company's business data.

Conclusion

This experiment helped in understanding how Power BI can be used to work with and analyze retail sales data. The Sample Superstore dataset with 9,994 records was imported from a CSV file and prepared using Power Query by removing unnecessary data and correcting the required data types. Different visuals were then created to study sales, profit, products, customers, and locations. The dashboard also includes slicers, which make it easier to view the data based on different regions and categories. Overall, the experiment showed how raw sales data can be converted into a simple and interactive dashboard for better business analysis.

Business Insights

The following key business insights were derived from the Superstore Sales Performance Dashboard:

- **Category Performance:** Technology generated the highest sales among the three main categories, showing that technology products have strong demand.
- **Customer Segment:** The Consumer segment contributes the largest share of sales, making it an important customer group for the business.
- **Sales Pattern:** Sales generally increased during the later years of the dataset, showing positive growth in overall business performance.
- **State-wise Sales:** California recorded the highest sales among the states, while lower-performing areas could be considered for future business expansion.
- **Overall Performance:** The dashboard recorded around 5.01K unique orders and \$2.30M in sales, indicating substantial customer activity during the period.
- **Interactive Filtering:** The Region and Category slicers make it possible to compare different parts of the business and quickly identify changes in sales performance.